Digital Twin Anchored Generation: Adaptive Querying and Phenotypic Refinement

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***Abstract*—Digital Twin Anchored Generation (DTAG) couples a language interface with Large Science Models (LSMs) of survey-response dependencies. Conditional predictions for unasked items define a *predictive phenotype*, refined by successive answers; geometric adaptive querying selects questions by expected movement of a pole-relative index. We distinguish such inference from *phenotypic construction*, in which interaction changes future reports or beliefs. Native GSS 2024 simulations with six pinned personas yielded 960 adaptive 20-question trajectories and 2,880 fixed-item order replays. Right- and left-derived question sets retained their directional shifts under original, reversed, and shuffled fixed-order replays, while neutral-derived sets remained near zero; no mean ordering effect was detected. These native-model experiments do not establish human persuasion or validate language mapping. A prospective randomized study will test whether adaptive questions move the estimated ideology in the predicted direction when answered by real respondents.**

***Index Terms*—Computational theory of mind, opinion dynamics, large science models, digital twins, survey methodology, social influence, polarization, latent geometry, adaptive questioning.**

I. Introduction

Belief systems comprise dependencies among economic, religious, institutional, and political attitudes, not only individual responses. Classical studies of issue constraint [1], [2] and later work on partisan sorting and alignment [3], [4] motivate this high-dimensional view. We ask whether sequential questioning reveals or helps construct such a system.

DTAG links natural-language queries to an LSM of empirical conditional survey responses. Each answer anchors a predictive phenotype, updates predictions for unasked questions, and moves a pole-relative geometric index. Adaptive queries select items by expected displacement, making sequential behavior measurable rather than merely plausible prose.

Synthetic ideological movement is not persuasion. We formalize *passive refinement* of fixed beliefs versus *reflexive construction* through interaction, then specify human tests of whether learned conditional geometry predicts more than inference.

Working research draft, October 2026. University of Kentucky. DTAG computations and human-subject studies described as proposed must not be interpreted as observed results.

II. Prior Work and the Scientific Gap

A. Belief systems, cognitive organization, and survey response

Converse [1] identified weak issue constraint outside politically sophisticated groups; Zaller [2] modeled reports as samples from accessible considerations. Later studies separate partisan sorting from issue alignment [3] and show how homophily and influence couple otherwise weak affinities [4]. These motivate measuring conditional attitude organization rather than assuming a single axis.

Consistency pressures [5], survey wording and order [6], and motivated evaluation [7] can change reports. Temporary salience, scale artifacts, reconstruction, and lasting revision require separate tests of persistence and generalization.

B. Quantitative influence and experimental evidence

DeGroot [8], Friedkin–Johnsen [9], and bounded-confidence [10] models specify social transitions but typically use low-dimensional individual states. DTAG instead models high-dimensional conditional structure without yet identifying a causal human transition operator.

Bail *et al.* [11] randomized incentives to follow Twitter bots providing cross-cutting political content. Republicans assigned a liberal bot later reported more conservative views; the corresponding Democratic estimate was not significant. This asymmetry motivates, but does not validate, DTAG. Campaign-contact experiments found small general-election mean effects [12], whereas some structured conversations produced more persistent reported changes [13]. Prediction must address heterogeneity and persistence.

C. Theory of mind and generative simulation

Theory of mind infers unobserved mental states from behavior [14], [15]; inverse planning formalizes this inference [16]. DTAG tests a narrower, falsifiable task: predict distributions of *unasked attitudes* from observed answers. Synthetic LLM personas [17], [18] may reproduce profiles without calibrated individual conditionals; DTAG exposes states, probabilities, and response provenance.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ range over $\mathcal{X} = \prod_i \mathcal{A}_i$, with finite response alphabets $\mathcal{A}_i$ and survey population, wave, or context c. An LSM is the conditional family

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated using coupled conditional inference trees [19]–[21]. Edges $j \to i$ indicate prediction, not causation; separately fitted conditionals may be jointly incompatible [22].

The respondent state is a partial assignment $s = \{X_j = a_j : j \in S\}$. In $\widetilde{x}_{-v}(s)$, all unanchored coordinates remain missing (no modal completion); the native trees marginalize missing splits by subtree-mass-weighted branch mixtures. This is a tree-wise prediction rule, not necessarily marginalization under one coherent joint law. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

B. Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{4}$$

Here D_{JS} is Jensen–Shannon divergence and the native code evaluates *qdistance*. Predictively indistinguishable states may be at zero distance: this is generally a pseudometric of *predictive consequence*, not a postulated Euclidean manifold of minds.

With 80% responses masked [21], mean LSM-distance distortion declined about 54.9% (GSS) and 60.9% (Eurobarometer). These are geometric reconstruction gains, not exact-label accuracy or causal influence.

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{5}$$

The semantically defined poles [21] specify 27 opposing GSS 2024 responses on social policy, civil liberties, public spending, and religion (Table I); all other coordinates remain unspecified. They are reference configurations, not typical respondents. The pole-relative index is not a self-reported ideology score.

IV. DTAG: A Conditional Theory-of-Mind Interface

A. Architecture and generative semantics

A semantic interface maps question q to survey variables $M(q)$. The LSM predicts responses, the answer anchors the state, and an LLM verbalizes it. For draw mode,

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \qquad v_t \in M(q_t), \tag{6}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{7}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{8}$$

In *draw mode*, an answer is sampled, not chosen by the optimizer; *pinned* anchors remain fixed. The system also supports modal answering, validated/multi-item mapping, and finite-memory *eviction* of old anchors, none used in the reported native runs. The fixed LSM conditions responses; it does not identify persuasion.

Figure 1 combines the architecture and native experiment.

B. Sequential geometry and adaptive questioning

For every candidate item v and response a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{9}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v) \Delta_v(a; s), \tag{10}$$

$$\sigma_v^2(s) = \sum_{a} p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{11}$$

Adaptive policies optimize $\mu_v(s)$ under conditional sampling; modal-response objectives instead use the most probable answer. A modal sequence is not a sampled respondent population. These displacements are model-relative, not proven human interventions.

The predictive change at unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \tag{12}$$

A weighted score $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$, with prespecified $w_j \geq 0$, quantifies predictive redistribution, not causal propagation.

C. Difference from RAG and language-model personas

RAG provides grounding text [23]; profile-prompted LLM personas seek plausible or distributionally faithful responses [17], [18]. DTAG instead exposes a native conditional model whose predictions for unasked responses can be calibrated as anchors accumulate. It complements, rather than replaces, text retrieval.

V. The Interpretation of Sequential Questions

A. Phenotypic refinement: observing a fixed system

Under passive refinement, respondents possess fixed phenotypes $X^\star$ and answers increasingly constrain predictions along $s_0 \subset s_1 \subset \cdots \subset s_T$. For target items Y, the information value of query v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v \mid s}[H(Y \mid s, v = A_v)]. \tag{13}$$

For a compatible joint law this is nonnegative conditional mutual information, distinct from geometric displacement μ_v in (10). Refinement can be tested by revealing held-out archival answers.

B. Phenotypic construction: querying a reflexive system

Under construction, response A_t to query q may alter latent state Z_t:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \tag{14}$$

TABLE I

GSS 2024 left/right reference poles. The 27 listed item–response assignments are the complete nonmissing coordinates of the two pole states used to compute I_Φ. All other survey variables are unspecified. Topics paraphrase item wording; categorical responses match the simulation definitions. The poles yield $I_\Phi = -1$ and $+1$.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

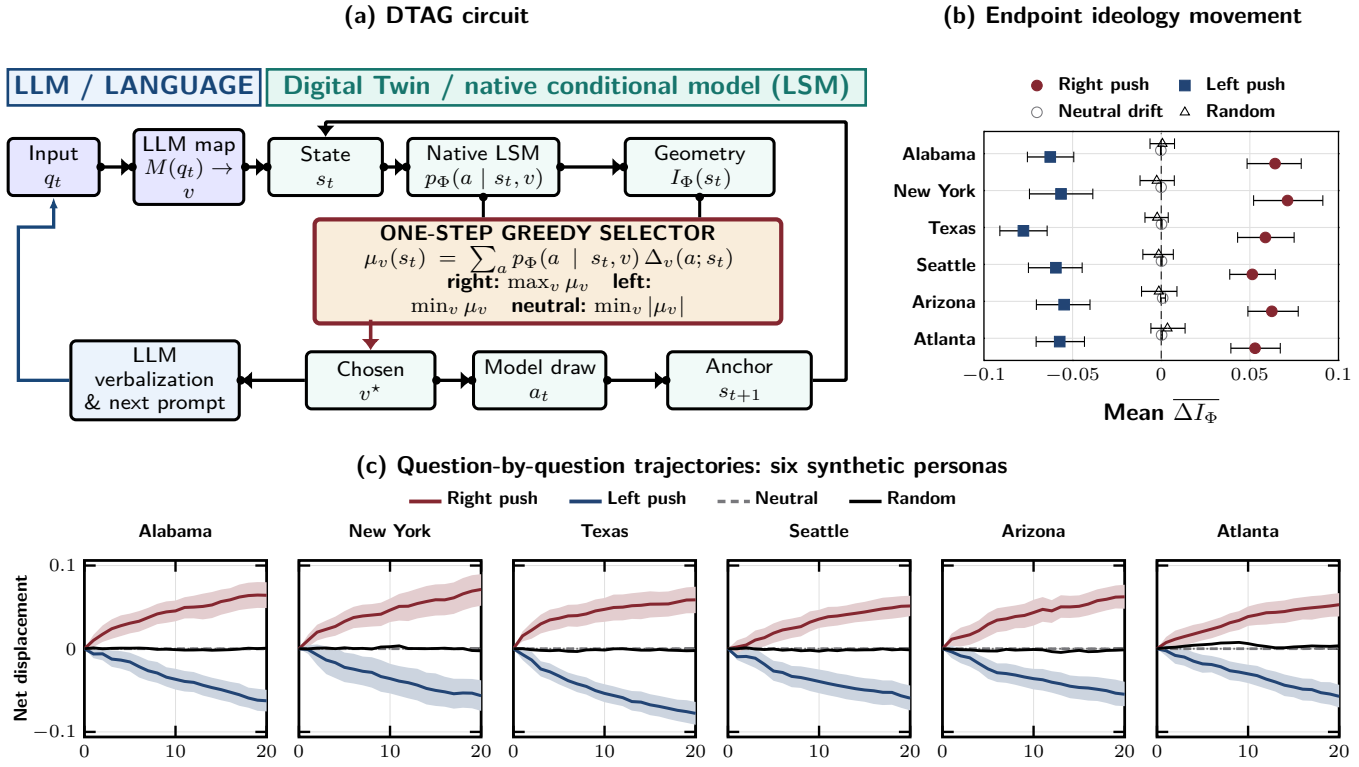


Fig. 1. **DTAG digital-twin circuit and adaptive native GSS 2024 results.** (a) A semantic LLM interface maps and verbalizes categorical survey questions, while the conditional LSM inside the digital twin predicts unasked responses; a myopic greedy circuit selects the next question using expected one-step pole-relative displacement and updates the anchored state with a model draw. (b) Mean displacement after 20 questions across six synthetic personas; large markers and horizontal whiskers show 95% bootstrap uncertainty. (c) Six aligned trajectories from 40 repetitions per persona and policy; shading denotes pointwise 95% bootstrap intervals. Right = dark red, left = dark blue, neutral = dashed gray, random = black. These are native conditional-model trajectories, not observed human persuasion.

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; otherwise transitions depend on prior beliefs, framing, trust, or history. Cross-sectional LSM conditionals cannot identify T_q, even with perfect observational fit.

Clarification, temporary salience, and lasting revision call for independent outcomes, delayed measurement, and no-query controls. We test whether statistical geometry predicts responsiveness as well as reconstruction.

C. Why question-order effects are not automatically persuasion

A coherent static joint model implies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \tag{15}$$

With a coherent joint law, terminal predictions after both answers are order-invariant, though intermediate states differ. Incompatible local conditionals can create order artifacts (15) [22]; genuine human operators K_A, K_B may

TABLE II
SIX PINNED SYNTHETIC PERSONAS IN THE NATIVE GSS 2024 SIMULATIONS.

Persona	**Pinned description**	**Initial $I_\Phi(s_0)$**
Alabama	45-year-old white man living in open country outside a small town in rural Alabama, in the South. He is married and has two children. He works full time in farming. He is a veteran who served 2 to 4 years of active duty in the armed forces. He reads a newspaper every day. His education ended with a high school diploma. He thinks of himself as conservative and is a strong Republican. He is Protestant and attends religious services every week.	0.075706
New York	22-year-old white woman living in New York City, a large central city in the Northeast. She has never been married and has no children. She works full time in retail. She has never served in the armed forces. She reads a newspaper every day. She has a bachelor's degree. She thinks of herself as extremely liberal and is a strong Democrat. She has no religion and never attends religious services.	−0.163807
Texas	61-year-old white woman in a small town in Texas, in the South. She is married with three children and keeps house. Her education ended with high school. She thinks of herself as extremely conservative and is a strong Republican. She is Protestant and attends religious services several times a week. She reads a newspaper every day.	0.045379
Seattle	34-year-old white man living in Seattle, a large central city in the West. He has never been married and has no children. He works full time and has a graduate degree. He thinks of himself as liberal and is a Democrat, but not a very strong one. He has no religion and never attends religious services. He reads a newspaper a few times a week.	−0.119639
Arizona	39-year-old Mexican American man living in a medium-size city in Arizona, in the West. He is married with three children and works full time. His education ended with high school. He thinks of himself as conservative and is a Republican, but not a very strong one. He is Catholic and attends religious services nearly every week.	−0.020437
Atlanta	47-year-old Black woman living in Atlanta, a large central city in the South. She is divorced with two children and works full time. She has a bachelor's degree. She thinks of herself as liberal and is a strong Democrat. She is Protestant and attends religious services every week. She reads a newspaper every day.	−0.030807

Persona descriptions and initial pole-relative ideology index $I_\Phi(s_0)$. Positive values point toward the designated right pole and negative values toward the left pole.

fail to commute. Simulation alone cannot distinguish them.

VI. Native GSS 2024 Computational Experiments

A. Adaptive questioning of fixed synthetic personas

We used six fixed synthetic GSS 2024 states (Table II), querying survey items directly without running LLM question mapping or verbalization. The native optimizer maximized expected rightward drift, minimized it for leftward drift, minimized absolute drift (neutral), or sampled eligible items randomly. Answers were drawn from LSM conditionals. At $T = 20$ and 40 repetitions per persona/policy, $6 \times 4 \times 40 = 960$ trajectories contain 19,200 draws.

Right- and left-seeking policies moved the index in their expected directions across all six personas; neutral minimized *geometric drift*, not psychological sensitivity.

B. Question selection as observed paths

Each observed question was assigned a shared categorical index without reordering runs or constructing modal paths. Of 228 items across all policies, 161 appeared in the 18 displayed right/left/neutral heatmaps (six personas by three policies; 40 runs of 20 questions). The global index permutation approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \left|\pi(v_{purt}) - \pi(v_{pur't})\right|, \tag{16}$$

Here u ranges over displayed policies. The all-pairs objective is independent of run numbering. The 67 random-only items remain in the lookup; index distances have no scientific meaning.

C. Fixed-set replay: robustness to question ordering

For each of six personas and three source policies (right, left, neutral), we froze the 20 questions from adaptive replicate zero without selecting on outcomes. Each set was replayed in its original, reversed, and two shuffled orders, with 40 fresh conditional draws per order ($18 \times 4 \times 40 =$ 2,880 runs). Question-specific uniform draws were coupled across orders to reduce sampling variability.

Figure 2 shows baseline-relative displacement for the reordered fixed sets. Directional movement persists: averaged across the six persona means, right-derived sets yield +0.0381 (original) and +0.0377 (reversed), while left-derived sets yield −0.0399 and −0.0410. For both original and reversed replays, all six persona-specific pointwise 95% bootstrap intervals exclude zero in the intended direction; both shuffles give the same directional pattern, and neutral-derived drift remains negligible.

All 54 pointwise bootstrap intervals for *paired mean between-order contrasts* include zero. Thus we find robust effects of the *selected question sets*, not evidence that permutation systematically changes the mean endpoint. These fixed-set replays are distinct from the continuously reoptimized trajectories of Fig. 1 and do not establish human belief change.

a) Cross-environment limitation.: Historical ten-question local and CI runs yielded different initial indices

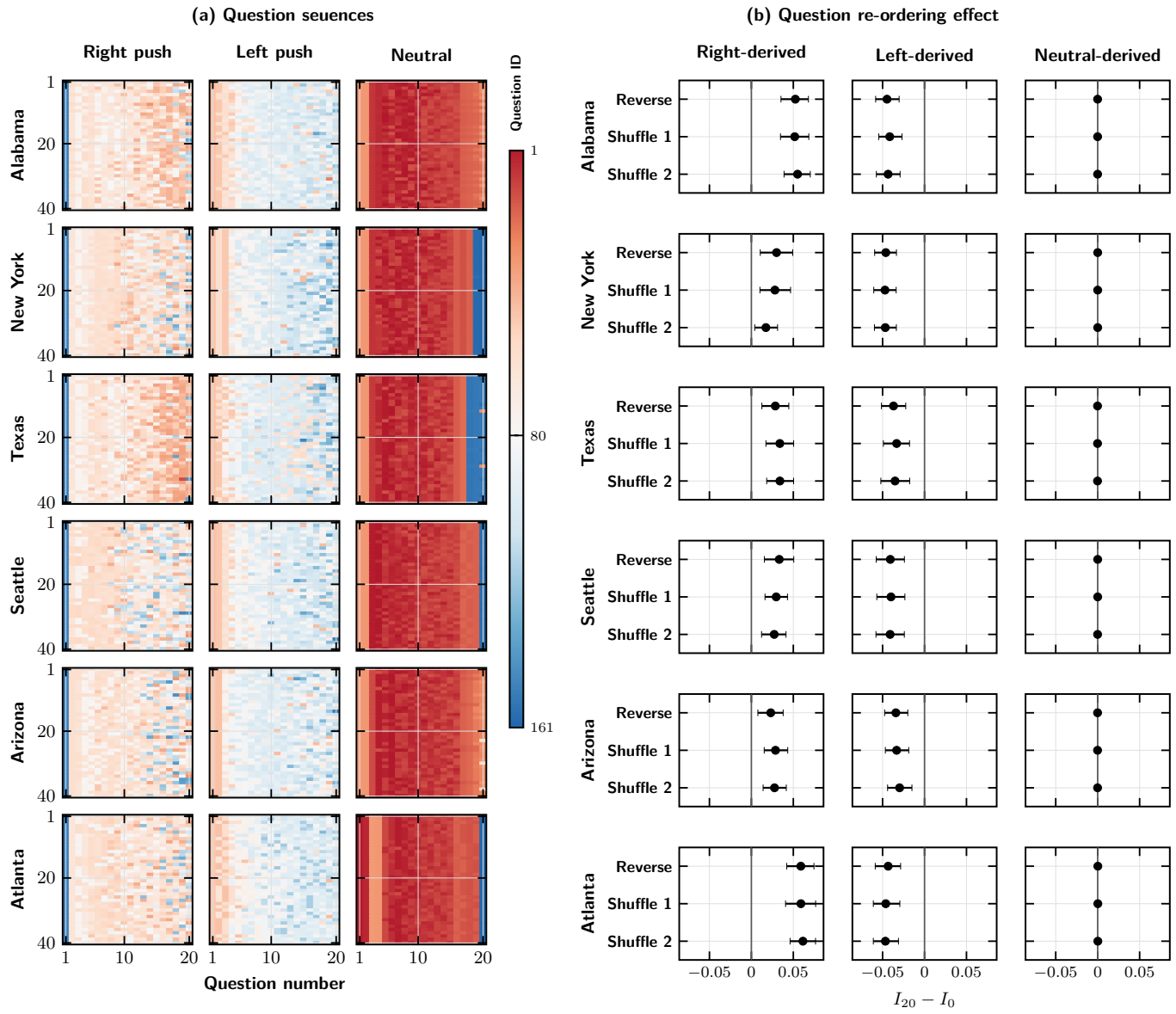


Fig. 2. **Question identities and fixed-item directional displacement across six personas.** Left: 18 categorical heatmaps show 40 runs over 20 question positions, with a shared colorbar identifying questions rather than ideological direction. Right: baseline-relative displacement $I_{20} - I_0$ and 95% bootstrap intervals for fixed optimized question sets replayed in reverse and two shuffled orders (40 repetitions each). Right- and left-derived sets retain directional movement; neutral-derived sets remain near zero. These are fixed-set replays, not the fully adaptive runs in Fig. 1.

despite matching persona, semantic-map, and pole-file hashes. Model binaries were not hashed in both manifests, so absolute cross-environment comparison requires runtime/model parity checks. Within-study results remain consistent.

VII. Testable Hypotheses and Prospective Validation

H1 (predictive refinement). The LSM's conditional response distributions should predict unasked answers from an initial set s_0, with improving held-out predictive scores as actual answers accumulate. Prior LSM reconstruction results motivate this assumption; prospective calibration is a supporting check, not the principal DTAG intervention test.

H2 (directional adaptive querying). After the same initial question sequence establishes $I_\Phi(s_0)$, successive questions selected by the one-step objective $\mu_v(s)$ should produce rightward or leftward shifts in I_Φ computed from respondents' *actual* answers, relative to neutral and random questioning. The direction of displacement, not a change in latent belief, is the primary falsifiable prediction.

a) Future validation on Prolific.: We propose a randomized study, not yet conducted, with a common baseline questionnaire and four question-selection arms: rightward, leftward, neutral, and random. The main comparisons test whether baseline-to-endpoint index displacement is positive under rightward selection and negative under leftward selection, beyond the control arms, using preregistered questions, policies, and confidence intervals. Predictive

probabilities will be recorded before each answer to assess calibration; independent unasked items and an optional delayed follow-up may explore whether changes extend beyond the fitted index. Sample size will follow pilot-based power analysis. Prolific permits recontact [24]; human-subjects oversight and safeguards against individually targeted political influence are required. This is a test of model-guided *estimation*, not an intervention designed to persuade voters. Order-specific human effects are secondary future questions, not required arms of this initial study.

VIII. Discussion and Limitations

DTAG models *what a respondent might report*, not yet *what an interaction makes them believe*. Associations can reflect common causes, sorting, survey design, or history. Incompatible LSM conditionals or semantic mappings may create synthetic order effects unrelated to human reflexivity.

Under passive observation of a coherent joint law, the posterior mean of a fixed trait $g(X)$ is a martingale. DTAG's nonlinear index $I_\Phi(s)$ is generally not that posterior mean and may drift without persuasion. Limitations also include pole and wave dependence, missingness calibration, wording, demand, response modes, and eviction of old anchors. Human experiments need measurement invariance and delayed outcomes.

IX. Conclusion

DTAG converts learned conditional survey dependencies into an operational question-selection policy that systematically shifts estimated ideology in native simulations. The shift persists when selected question sets are replayed in different orders. The central prospective test is whether the policy predicts the direction of estimated-ideology movement from actual human answers, without equating such movement with persuasion.

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